

Surname: _____

Given Name: _____

Student Number: _____

FINAL EXAMINATION

December 2022 Final Exams

ELECENG 3TP3

Instructors: T. Todd and T. Kirubarajan

YOU ARE RESPONSIBLE FOR ENSURING THAT YOUR COPY OF THE PAPER IS COMPLETE. BRING ANY DISCREPANCY TO THE ATTENTION OF YOUR INVIGILATOR.

Examination Duration: 150 minutes**Exam Instructions:**

SINGLE VERSION exam

Materials Permitted In The Exam Venue:

No electronic aids are permitted, e.g., laptops, phones

McMaster Standard Calculator (Casio FX-991MS/MS+)

One Double Sided 8.5" x 11" Crib Sheet

Materials To Be Supplied To Students:

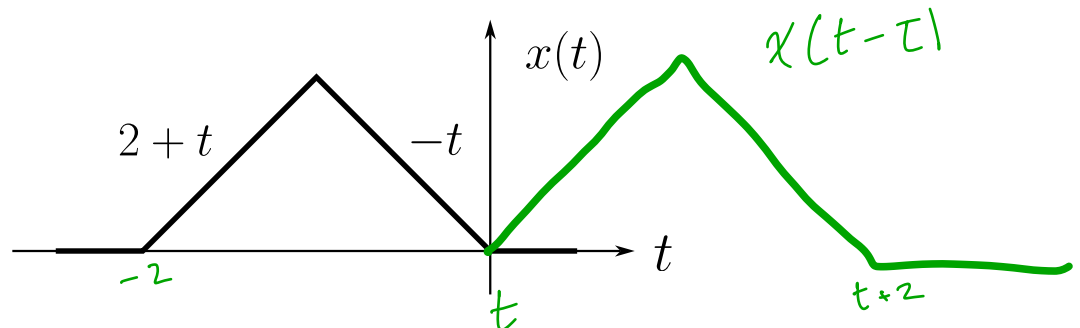
None

Instructions To Students:

Answer all questions on the sheets provided.

Q1	Q2	Q3	Q4

- (25) 1. Consider the single triangular pulse $x(t)$ shown below. Define $y(t) = x(t) * x(t)$ as the convolution of $x(t)$ with itself.



- (a) If $x(t - \tau)$ is the impulse response of a causal linear time invariant system, what values of τ would be permitted?

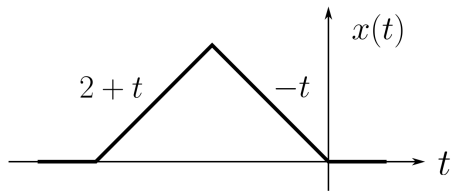
$$\tau \geq 2$$

- (b) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-4 \leq t \leq -3$. Do not evaluate the integral.

$$\int_{-2}^{t+2} (2+\tau)(2+t-\tau) d\tau$$

- (c) Write out the integral (including the integration limits) that can be evaluated in order to find $y(t)$ for $-2 \leq t \leq -1$. Do not evaluate the integral.

- 1) 1. Consider the single triangular pulse $x(t)$ shown below. Define $y(t) = x(t) * x(t)$ as the convolution of $x(t)$ with itself.



- (a) If $x(t - \tau)$ is the impulse response of a causal linear time invariant system, what values of τ would be permitted?

a)

$$\int_t^{-1} (2 + \tau)(\tau - t) d\tau + \int_{-1}^{t+1} (\tau - t)(-\tau) d\tau + \int_{t+1}^0 (-\tau)(2 + t - \tau) d\tau$$

(25) 2. Answer the following questions relating to discrete time signals and systems.

(a) Find a closed form expression for the convolution, $h[n] * f[n]$, where

$$h[n] = 2^{-n}u[n] - \delta[n - 1]$$

$$f[n] = 2^{-n}u[n]$$

Best to treat the impulse separately, i.e., $h[n] = g[n] + \delta[n - 1]$, therefore, $h[n] * f[n] = g[n] * g[n] - 2^{-(n-1)}u[n - 1]$. Now $g[n] * g[n] = \sum_{i=-\infty}^{\infty} g[i]g[n - i] = \sum_{i=0}^n 2^{-n} = (n + 1)2^{-n}u[n]$. Final answer: $h[n] * f[n] = (n + 1)2^{-n}u[n] - 2^{-(n-1)}u[n - 1]$

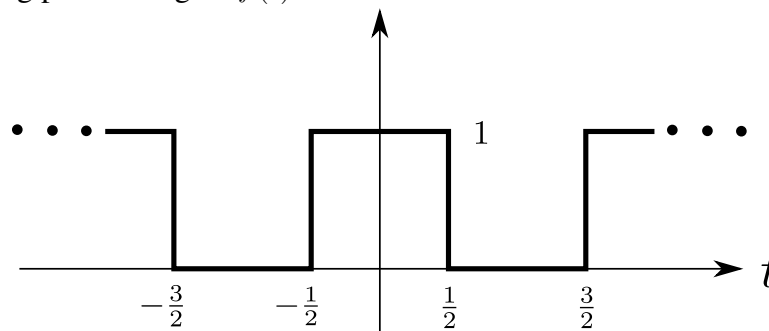
(b) Find the first five non-zero values of the impulse response of the system defined by

$$y[n] - y[n - 1] = x[n] + x[n - 1]$$

where $x[n]$ and $y[n]$ are the system input and output, respectively.

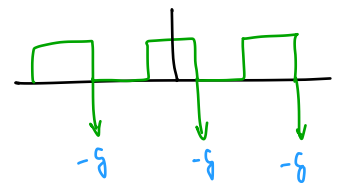
starting at $n = 0$ we have 1, 2, 2, 2, 2.

(25) 3. The following periodic signal $f(t)$

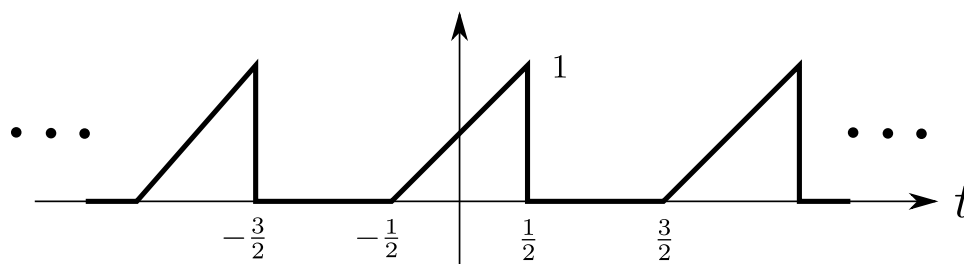


has complex exponential Fourier series coefficients given by

$$c_k = \frac{\sin(k\pi/2)}{k\pi} = \frac{\text{sinc}(k/2)}{2} \quad \text{for } k = 0, \pm 1, \pm 3, \dots$$



(a) Consider the following periodic signal, $d(t)$.



$$2) \quad x[n] = \delta[n]$$

$$y[n] = y[n-1] + \delta[n] + \delta[n+1]$$

$$y[0] = 0 + 1 + 0 = 1$$

$$y[1] = 1 + 0 + 1 = 2$$

$$y[2] =$$

$\vdots$

3)

Use the relationship between $f(t)$ and $d(t)$ to obtain the complex exponential Fourier series coefficients, d_k , of $d(t)$.

Note that $d'(t) = f(t) - \sum_{n=-\infty}^{\infty} \delta(t - 1/2 + 2n) \leftrightarrow e_k$. **Thus** $e_k = c_k - e^{-jk\pi/2}/2$.
Therefore $d_k = e_k/jk\pi$ (**for** $k \neq 0$), **i.e.**, $d_k = \frac{\text{sinc}(k/2)}{j2k\pi} - e^{-jk\pi/2}/2jk\pi$. $d_0 = 1/4$.

- (b) Find a value for the average power in the signal $d(t)$ accurate to two decimal places.

Obtained as an integral in time. $\frac{1}{2} \int_{-1/2}^{1/2} (t + 1/2)^2 dt = 1/6$. ** can also use Parseval's theorem w/ Fourier coeff.s but more difficult*

- (c) Assume that $f(t)$ is input to a linear time-invariant (LTI) system whose impulse response has the following Fourier transform.

$$H(\omega) = \frac{\pi}{j\omega + \pi}.$$

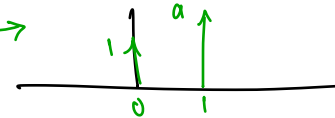
Find the phase shift (in degrees) that the fundamental frequency component in $f(t)$ will experience.

The fundamental frequency is $\omega_0 = \pi$. **Therefore** $H(\pi) = \frac{1}{j+1}$. **The phase shift is therefore** $-\tan^{-1}(1) = -45$ **degrees.**

- (25) 4. A discrete-time model of a two-path wireless propagation channel has an impulse response given by $h[n]$ as follows.

$$h[n] = \delta[n] + a\delta[n-1]$$

where a is a constant.

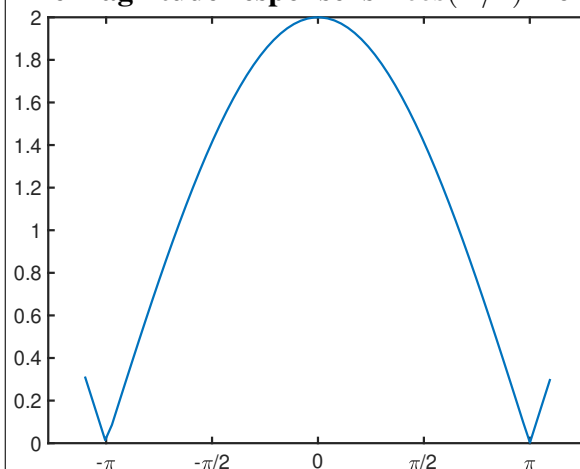


- (a) Find the discrete-time Fourier transform, $H(\Omega)$, of $h[n]$ when $a = 1$. Simplify the expression as much as possible.

$H(\Omega) = 1 + ae^{-j\Omega}$. **When** $a = 1$, $H(\Omega) = 1 + e^{-j\Omega} = e^{-j\Omega/2}(e^{j\Omega/2} + e^{-j\Omega/2}) = 2e^{-j\Omega/2} \cos(\Omega/2)$. *factor out $e^{-j\Omega/2}$*

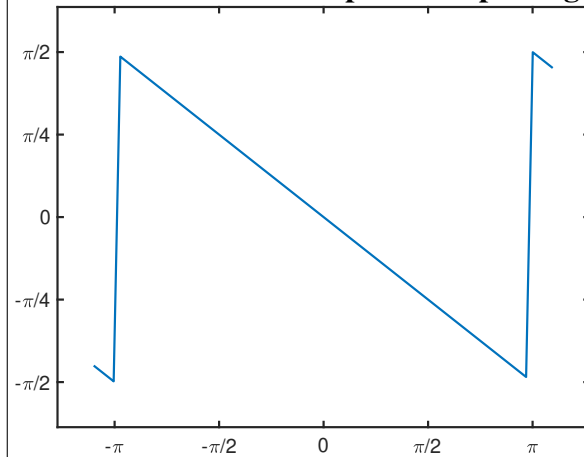
- (b) Find and sketch the magnitude of $H(\Omega)$ for Ω in the range $[-\pi, \pi]$.

The magnitude response is $2 \cos(\Omega/2)$ **from** $[-\pi, \pi]$.



- (c) Find and sketch the phase of $H(\Omega)$ for Ω in the range $[-\pi, \pi]$.

The phase response over $[-\pi, \pi]$ is given by the phase of $e^{-j\Omega/2}$, i.e., $-\Omega/2$. It is a saw-tooth function with a slope of $-1/2$ passing through $\Omega = 0$. It then repeats continually.



END OF EXAMINATION